BrainyEdHub
The Hub for Smarter Education.

MOCK EXAMINATION — TRY BEFORE YOU BOOK

GCSE Statistics

Mock Test & Mark Scheme

Time allowed	25 minutes
Total marks	39 marks
Number of questions	27
Topics covered	Data & Sampling, Averages & Spread, Representing Data, Probability

Student name: ______________________ **Date:** ______________________

Instructions

- Answer every question in the space provided on your own paper.
- Write your name and today's date at the top of your answer sheet.
- The marks for each question are shown in brackets, e.g. **(2 marks)**.
- Where a calculation is required, show your working — marks are available for method as well as the final answer.
- Once you finish, check your answers against the mark scheme at the back of this document.
- This mock test is provided free by BrainyEdHub to give you a feel for our teaching standard — book a free trial class to go through it with a tutor.

Question Paper

Data & Sampling

1. Give the term for data that can only take exact, countable values (e.g. number of siblings). **[1 mark]**

2. Give the term for data that can take any value within a range (e.g. height). **[1 mark]**

3. A researcher surveys 50 out of 2000 students at a school. State the sampling fraction. **[1 mark]**

4. State one advantage of using a random sample rather than surveying an entire population. **[2 marks]**

5. Define 'population' in the context of statistics. **[1 mark]**

6. Give one example of a sampling method other than random sampling. **[1 mark]**

Averages & Spread

7. A gardener records plant heights (cm): 12, 14, 16, 18, 20. Find the mean. **[2 marks]**

8. Find the median of 7, 3, 9, 5, 11. **[2 marks]**

9. Find the mode of 2, 3, 3, 4, 5, 3, 6. **[1 mark]**

10. Find the range of 4, 9, 15, 3, 20. **[1 mark]**

11. Name one measure of spread other than the range. **[1 mark]**

12. Find the mean of 10, 15, 20, 25, 30. **[2 marks]**

13. A data set contains one very high value compared to the rest. State which average this outlier affects the most. **[1 mark]**

Representing Data

14. A grouped frequency table has class intervals 0–9 (frequency 3), 10–19 (frequency 7), and 20–29 (frequency 5). Find the total frequency. **[1 mark]**

15. Using the midpoints 4.5, 14.5 and 24.5 for the table above, estimate the mean. **[3 marks]**

16. State one reason a pie chart might be a poor choice for displaying data with 20 categories. **[2 marks]**

17. Name a suitable graph for displaying how a value changes over time. **[1 mark]**

18. Name a suitable graph for comparing the frequencies of separate categories. **[1 mark]**

19. State what a scatter graph is used to show. **[1 mark]**

20. In a scatter graph, what does a line of best fit help you do? **[2 marks]**

Probability

21. A bag contains 5 red, 3 blue and 2 green balls. Find P(blue). **[1 mark]**

22. Two fair dice are rolled and their scores added. Find P(the total is 7). **[3 marks]**

23. Explain the difference between correlation and causation, using an example. **[2 marks]**

24. A spinner has 4 equal sections numbered 1–4. Find P(a number greater than 2). **[1 mark]**

25. State the term used when two events cannot happen at the same time. **[1 mark]**

26. The probability that an event happens is 0.7. State the probability that it does not happen. **[1 mark]**

27. A coin is flipped twice. Find the probability of getting two heads. **[2 marks]**

Mark Scheme

Use this guide to check your answers. Award full marks for correct final answers; where working is shown, award partial credit as indicated.

Q	Model answer / mark scheme	Marks
1	**Discrete data**	1
2	**Continuous data**	1
3	**1/40 (2.5%)** 50 ÷ 2000 = 1/40.	1
4	**Any valid reason, e.g. it saves time and cost while still giving a representative result.**	2
5	**The entire group of people or items being studied.**	1
6	**E.g. stratified sampling, systematic sampling, or quota sampling.**	1
7	**16 cm** Sum = 80, 80 ÷ 5 = 16.	2
8	**7** Ordered: 3, 5, 7, 9, 11 — the middle value is 7.	2
9	**3**	1
10	**17** 20 – 3 = 17.	1
11	**Interquartile range (or standard deviation)**	1
12	**20** Sum = 100, 100 ÷ 5 = 20.	2
13	**The mean**	1
14	**15**	1
15	**15.83 (2 d.p.)** (3×4.5 + 7×14.5 + 5×24.5) ÷ 15 = 237.5 ÷ 15.	3
16	**Any valid reason, e.g. too many slices become hard to read and compare accurately.**	2
17	**A line graph**	1
18	**A bar chart**	1
19	**The relationship (correlation) between two variables.**	1
20	**It shows the trend of the relationship and can be used to estimate/predict other values.**	2
21	**3/10**	1

Q	Model answer / mark scheme	Marks
22	**1/6** 6 favourable outcomes (e.g. 1+6, 2+5, 3+4...) out of 36 possible outcomes.	3
23	**Correlation means two variables change together; causation means one directly causes the change in the other.** E.g. ice cream sales and sunburn cases are correlated, but ice cream doesn't cause sunburn — both are caused by hot weather.	2
24	**1/2**	1
25	**Mutually exclusive**	1
26	**0.3**	1
27	**1/4** P(H) × P(H) = ½ × ½.	2

Total for Statistics	**39 marks**

How did you do? A BrainyEdHub tutor can go through any question you got stuck on, and build a personalised revision plan around your results. Book a free trial class at **brainyedhub.com/contact.html** or message us on WhatsApp.